

Properties of crystalline solid

10/10/21

1. Long range orderly arrangement of constituents.

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2. Definite / regular shape

3. Anisotropic in nature

4. Definite value of heat of fusion

5. They have sharp melting point.

Amorphous solid

Amorphous solid is a non-crystalline solid in which the atoms and molecules are not organized in a definite lattice pattern. Example: Rubber, plastic, glass etc.



fig 02 : amorphous solid

Properties of amorphous solid

1. Short range, random arrangement of constituents.

2. Irregular shape

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3. Isotropic in nature

4. doesn't have definite value of heat of fusion.

5. doesn't have sharp melting point.

Polycrystalline solid

Polycrystalline solid is consistence of many crystalline parts that are randomly oriented with respect to each other. Example: All metals, rocks, ice and many ceramics



fig 03: polycrystalline solid

Properties of amorphous solid

1. Short range, random arrangement of constituents.
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Polycrystalline solid

Polycrystalline solid is consistence of many crystalline parts that are randomly oriented with respect to each other. Example: All metals, rocks, ice and many ceramics



fig 03: polycrystalline solid

Properties of polycrystalline solid

1. Random orientation of constituents

2. Isotropic in nature

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3. have a high degree of order over many atomic or molecular dimensions.

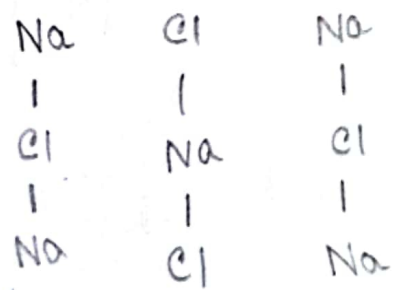
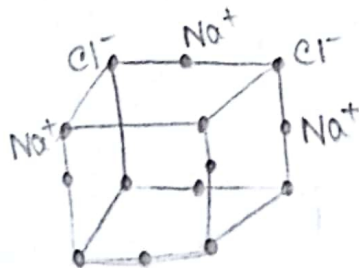
Crystal

1. Single crystal

2. Poly crystal

Single crystal

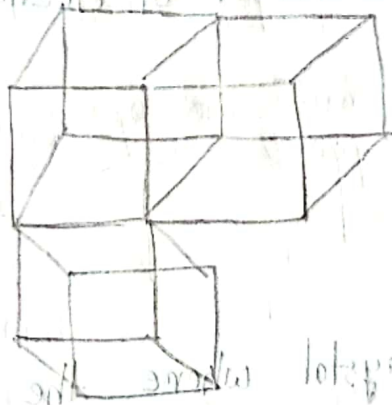
A crystalline solid in which the periodic and repeated arrangements of atoms is perfect or extends throughout entirely of the specimen without interruption is called single crystal. Example: NaCl, diamond



Poly crystal

A crystalline solid which one is composed of millions of single crystal or grains are called polycrystal or polycrystalline solid. It has regular shape. Example - metals, alloys, minerals, polymers.

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Ionic crystal

The crystal where lattice is made of by positive and negative ions which are held together by strong ionic bonds is called ionic crystal. In ionic crystal the valence electron are completely transferred from one atom to another.

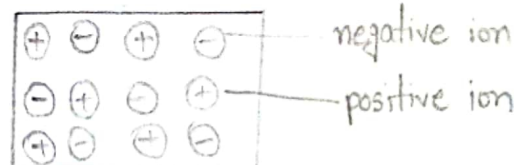


fig: Ionic crystal

Properties of ionic crystal

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1. High melting and boiling point
2. ^{soluble in} polar liquid
3. hard and brittle in nature
4. good conductor of electricity (NaCl, CsCl, CaCO_3 , MgCl_2 , LiF, MgSO_4)

covalent crystal

The whe crystal where the atoms of lattice are held together by sharing valence electron and form covalent bond is called covalent crystal. These crystals form a network and constructs a big molecule. Example: Graphite, SiO_2 , Diamond.

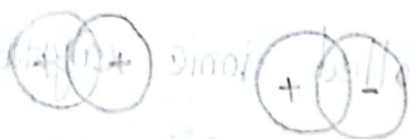


fig: covalent crystal

Properties of covalent crystal

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1. High melting and boiling point

2. Shows hardness

3. Non-volatile

4. Soluble in non polar solvent

5. Semiconductor type.

Molecular crystal

The crystal whose lattice is made by polar and non-polar gap and held together with vander wall force is called molecular crystal. Example: dry ice, solid forms of the noble gases and crystals of numerous organic compounds.

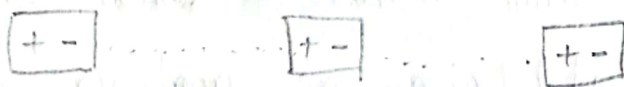


fig: molecular crystal

Properties of molecular crystal

1. Low melting and boiling point
2. Insoluble in polar liquid
3. Volatile in nature
4. Soft
5. Electrically insulator

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Metallic crystal

Crystal whose lattice has positive ion and surrounded by mobile electron is called metallic crystal. Example - Iron, Sodium, etc copper etc.

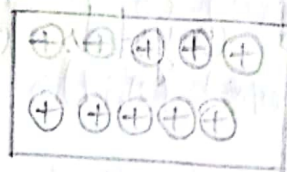


fig: metallic crystal

Properties of metallic crystal

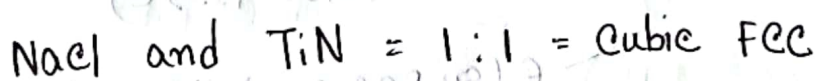
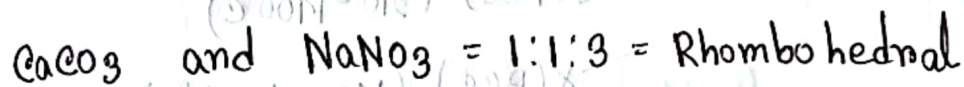
1. Hard or soft (depends on metal's property)
2. Melting and boiling point will vary
3. Can Both thermal and electrical conductor
4. Brittle and shiny.
5. Ductile.

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Isomorphism

Different chemical compounds that have same crystalline structure is called isomorphism. These compounds are called isomorph.

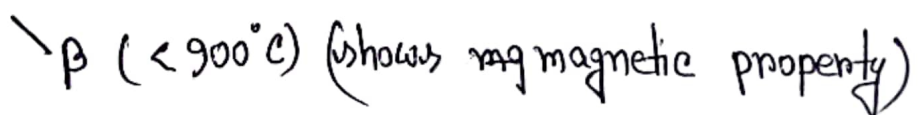
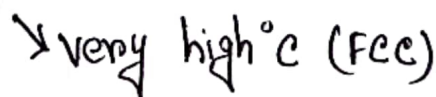
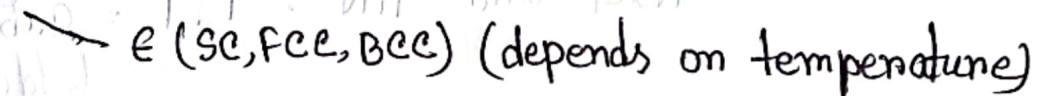
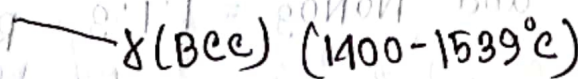
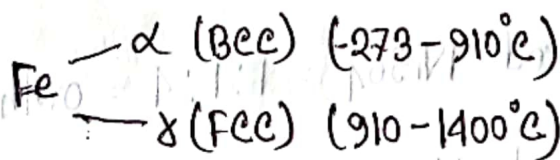
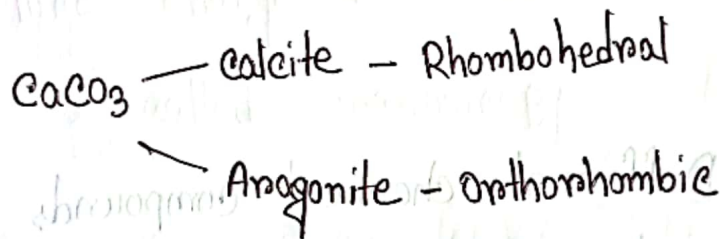
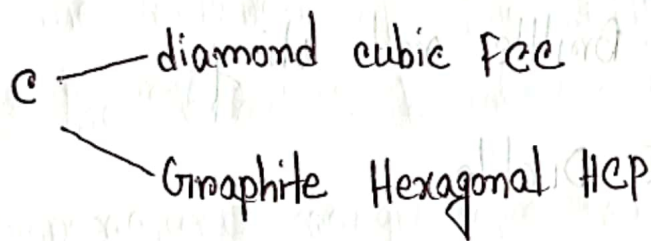
Example -



Polymorphism

Polymorphism is the ability of a solid material to exist more than one form or crystal structure.

Example:



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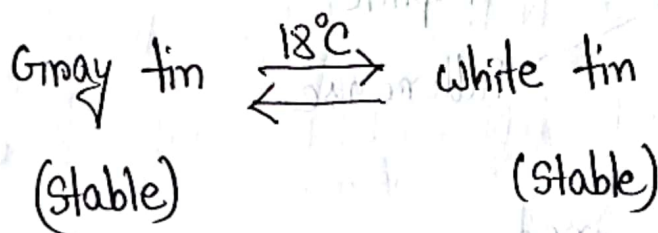
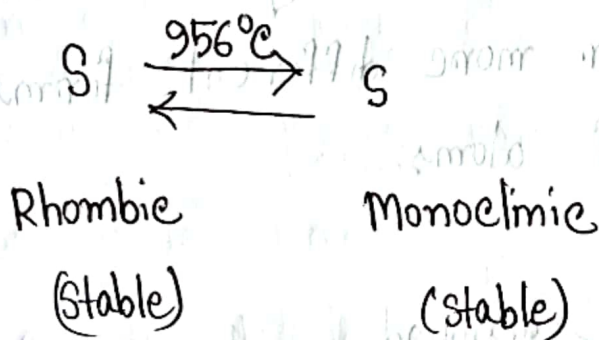
* All allotropic are polymorphism but all polymorphism are not allotropic.

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Enantiotropy

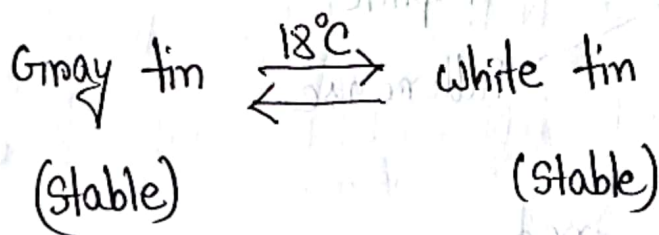
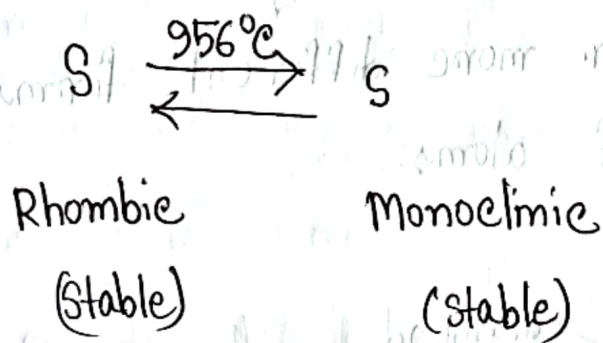
The relation of two different crystalline forms of the same substance (S, S_n) that have a definite temperature range of transition point and can therefore change reversible each other. This phenomena is called enantiotropy.



* All allotropic forms are polymorphs but all polymorphs are not allotropic.

Enantiotropy

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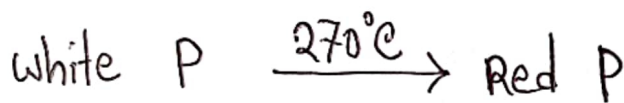


Monotropy

The relation of different crystalline form of the same substance (Ex: white Powdered P) that have no transition point since only one form is stable and the other form is unstable and the change unstable to stable form is irreversible. This phenomena is called monotropy.

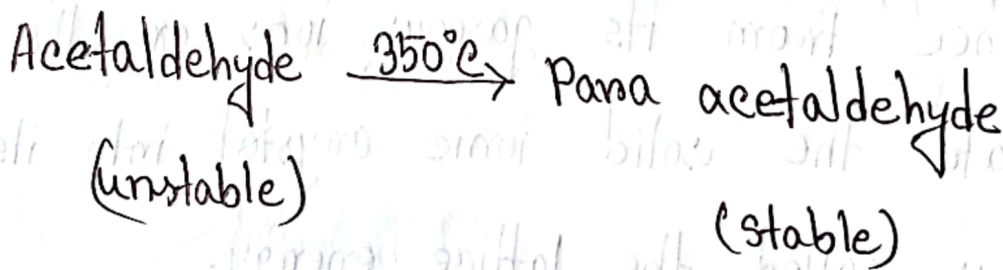
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(unstable)

(stable)



(unstable)

(stable)

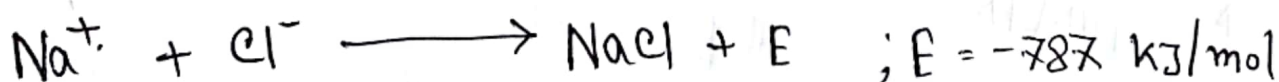
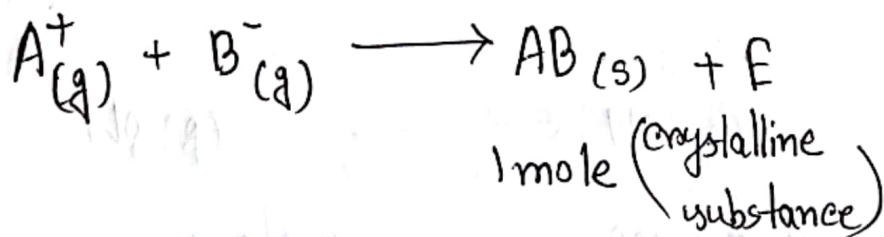
Factors that affects polymorphism

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1. Stability
2. Concentration
3. Growth of pattern
4. Temperature

Lattice energy

The amount of energy released during the formation of one mole of solid crystalline substance from its gaseous ions or the spent to separate the solid ionic crystal into its constituent ions is called the lattice energy.



Factors that affect lattice energy

1. **Size of ions**: If radius increases, lattice energy becomes less exothermic. This is because the charge on the ions is more spread out over the ion when ions are larger. Therefore the electrostatic forces of attraction between the oppositely charged ions in the lattice are weaker.

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2. **Charge of ions**: The lattice energy gets more exothermic as the ionic charge of the ions increases. The greater is the ionic charge, the higher is the charge density. This results in stronger electrostatic attraction. Thus the lattice energy is more exothermic.

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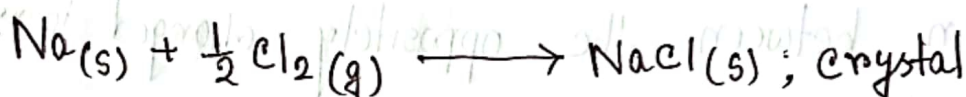
Born haber cycle

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1. Direct formation

2. Indirect formation

1. Direct formation:

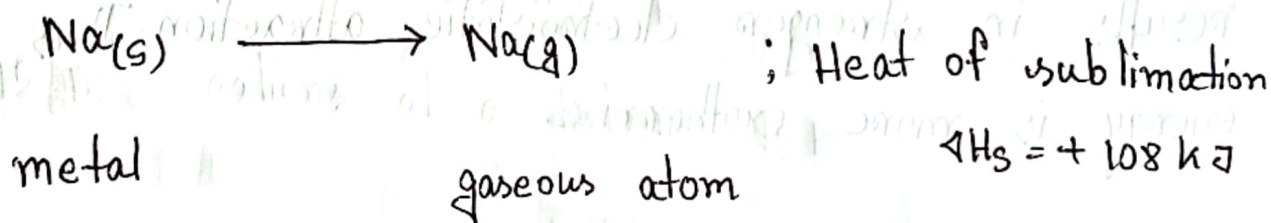


Heat of formation

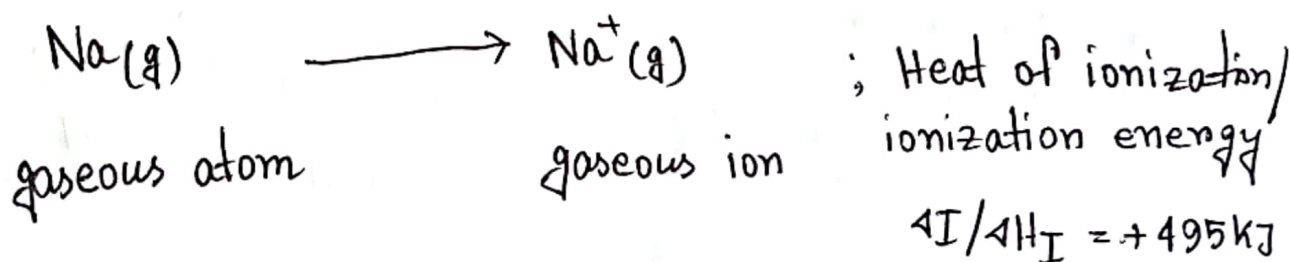
$$\Delta H_f = -411 \text{ kJ}$$

2. Indirect formation

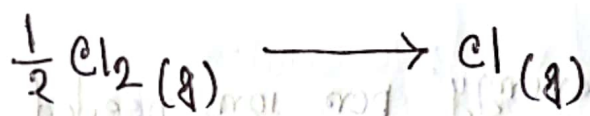
Step 1



Step 2



Step 3



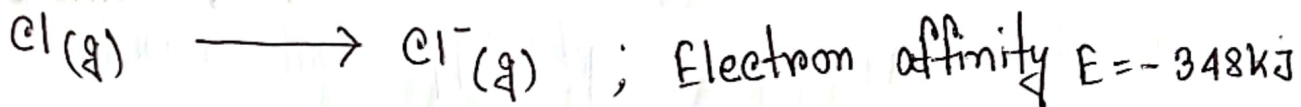
, Heat of dissociation

molecule

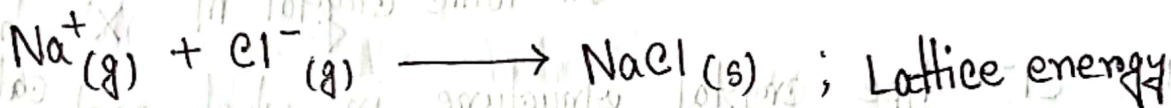
atom

$$\frac{1}{2} \Delta H_D = +121 \text{ kJ}$$

Step 4



Step 5

 $u = ?$

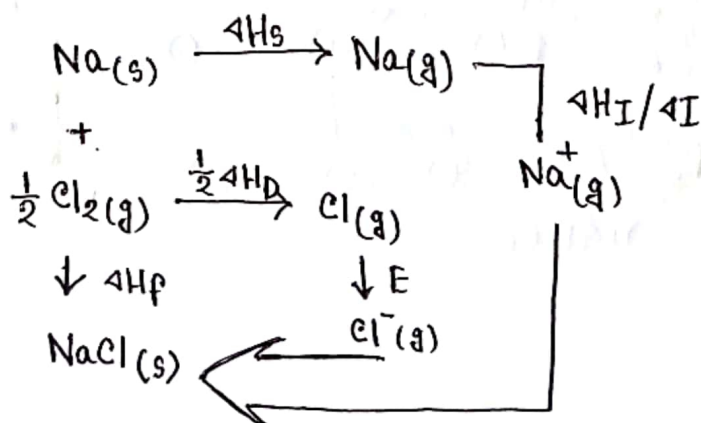
Hess's law

$$\Delta H_f = \Delta H_s + \Delta H_I + \frac{1}{2} \Delta H_D + E + u$$

$$\Rightarrow u = -787 \text{ kJ}$$

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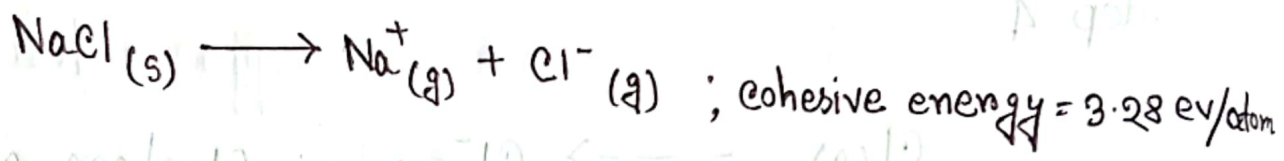
Here's the diagram -



Cohesive energy

The amount of energy per ion needed to break an ionic crystal into its constituent ions is called cohesive energy.

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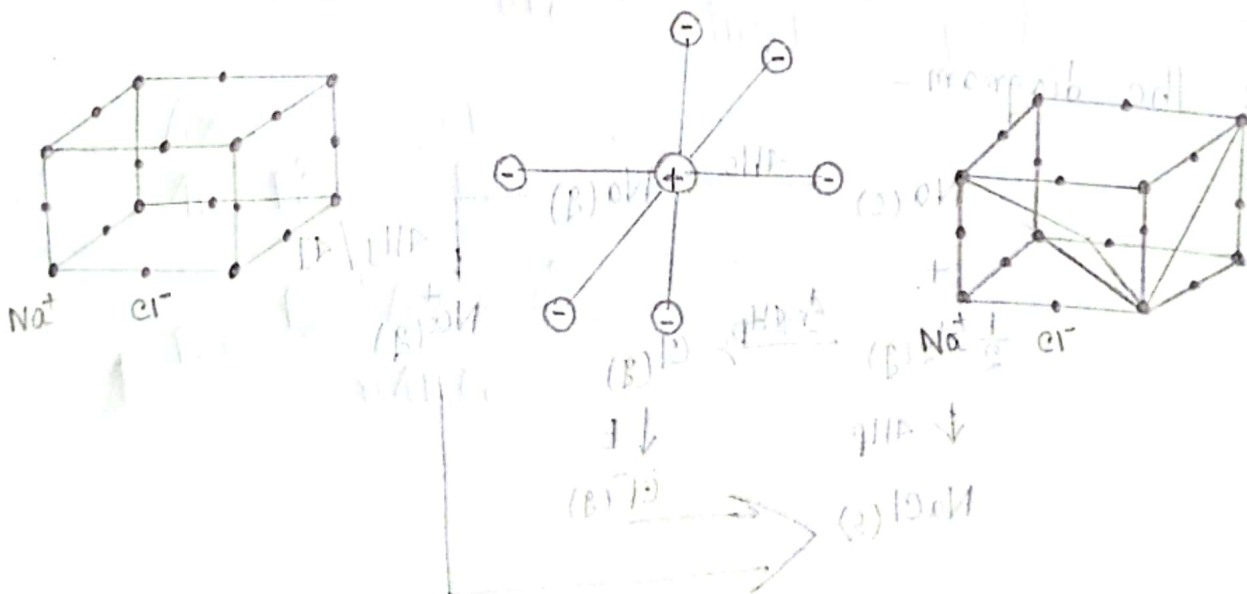


Expression

Let us consider an ionic crystal $M^{Z+} X^{Z-}$ having $\text{Na}^+ \text{Cl}^-$ type crystal structure in which each cation is surrounded by M^{Z+} cation X^{Z-} anion.

i) $6 X^{Z-}$ anions at a distance R

ii) $12 M^{Z+}$ cations at a distance $\sqrt{2} R$



Here,

B = Born-Land constant / repulsive constant

R = Ionic distance

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n = number of atom / Born exponent

Total potential energy,

$$U_{\text{total}} = U_{\text{attractive}} + U_{\text{repulsive}}$$

$$= \frac{-Az^+z^-e^2}{4\pi\epsilon_0 R} + \frac{B}{R^n} \quad \text{--- (iii)}$$

At equilibrium separation, R_0 of the ions, U is minimum and $\frac{dU}{dR} = 0$ when $R = R_0$

$$\therefore \left(\frac{dU}{dR} \right)_{R=R_0} = \frac{d}{dR_0} \left(\frac{-Az^+z^-e^2}{4\pi\epsilon_0 R} + \frac{B}{R^n} \right)$$

$$\Rightarrow \frac{Az^+z^-e^2}{4\pi\epsilon_0 R_0^2} - \frac{nB}{R_0^{n+1}} = 0$$

$$\Rightarrow B = \frac{R_0^{n+1} Az^+z^-e^2}{n4\pi\epsilon_0} \quad \text{--- (iv)}$$

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Substituting the value of B in equation (iii) will get the potential energy at the equilibrium separation,

$$U = \frac{-Az^+z^-e^2}{4\pi\epsilon_0 R_0} + \frac{R_0^{n-1} Az^+z^-e^2}{n4\pi\epsilon_0} \times \frac{1}{R_0}$$

$$= \frac{-Az^+z^-e^2}{4\pi\epsilon_0 R_0} + \frac{Az^+z^-e^2}{n4\pi\epsilon_0 R_0}$$

$$= \frac{-Az^+z^-e^2}{4\pi\epsilon_0 R_0} \left(1 - \frac{1}{n}\right)$$

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This is the Born land equation for cohesive energy of an ionic crystal.

$$\text{Lattice energy} = \text{cohesive energy} \times N_A$$

$$= \frac{-Az^+z^-e^2 N_A}{4\pi\epsilon_0 R_0} \left(1 - \frac{1}{n}\right)$$

* The madelung constant A for NaCl (Rock salt) is 1.7

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Solid Materials

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Solid

Solid is defined as a body ^{or} a geometric figure having three dimensions.

Types of solids - 1. Crystalline

2. Amorphous

3. Polycrystalline

Crystalline solid

A crystalline solid is one in which the atoms are situated in a repeating or periodic array over large atomic distances. Example: NaCl, diamond

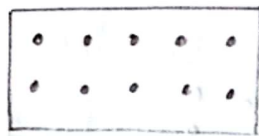


Fig 01: Crystalline solid